

Total No. of Questions : 8]

SEAT No. :

**PB2544**

[Total No. of Pages : 2

**[6263] 444**

**B.E. (Computer Engineering)(Honors in VR &AR)**  
**APPLICATION DEVELOPMENT USING AUGMENTED**  
**REALITY AND VIRTUAL REALITY**  
**(2019 Pattern) (Semester - VIII) (410703)**

**Time : 2½ Hours]**

**[Max. Marks : 70**

**Instructions to the candidates:**

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Make suitable assumptions if necessary.
- 4) Figures to the right indicate full marks.

- Q1)** a) What is ARCore? Explain the features of ARCore. [6]  
b) Explain: SLAM algorithm. [6]  
c) Write a note on AR tool kit. [5]

OR

- Q2)** a) Write difference between Environmental Detection and Occlusion. [6]  
b) Explain anatomy of SLAM algorithm. [6]  
c) What is Vuforia? Explain the features of Vuforia. [5]

- Q3)** a) What programming languages are commonly used for AR and VR development, what are the strengths and weaknesses of each of them? [6]  
b) Write syntax used for defining [6]  
i) Classes  
ii) Functions  
c) Discuss the importance of User experience in AR and VR application, and how programming languages can be used to enhance UX. [6]

OR

**P.T.O.**

- Q4)** a) Explain the role of unreal engine in AR and VR development. [6]  
b) What are some of the most popular technologies offer for AR and VR development? [6]  
c) Explain features of C# for AR and VR comfortable. [6]

- Q5)** a) What is HTC Vive? Explain the Working of HTC vive. [5]  
b) Explain the terms: [6]  
i) Tracking system  
ii) Monitoring system  
c) Explain: The contrast of hardware requirements for developing AR and VR applications. [6]

OR

- Q6)** a) Explain different types of AR systems. [4]  
b) What are the important components of AR system. [6]  
c) Explain Advantages and Disadvantages of AR and VR system. [7]

- Q7)** a) What are some trending application areas in AR and VR applications? [6]  
b) Explain the role of aerospace and defense in AR and VR. [6]  
c) What are some of the key areas of growth in AR and VR, and how are these trends likely to shape the future of the industry? [6]

OR

- Q8)** a) Evaluate the potential for AR and VR for healthcare applications. [6]  
b) Discuss the impact of AR and VR on entertainment industry. [6]  
c) Discuss the potential impact of AR and VR in education and learning. [6]

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